

FETAL HEALTH CLASSIFICATION WITH ARTIFICIAL NEURAL NETWORKS AND SUPPORT VECTOR MACHINES

ABSTRACT

Abnormal fetal heart rate patterns are key indicators of fetal health. The ability to classify fetal health states based on cardiotocographic (CTG) data is explored in this research with the use of artificial neural networks and support vector machines applied to target three classes. Adaptive synthetic sampling was used to address severe data imbalance and a two stage hyper-parameter search was applied. Performance of both models had varied strengths for different evaluation metrics, however the ANN was better in terms of sensitivity (0.88) and geometric mean (0.92) which is desired in the medical diagnostics context.

INTRODUCTION

The World Health Organization approximates that 6.3 million perinatal deaths occur every year worldwide with rates varying from 10 to over 30 deaths (per 1000 births) for developed and underdeveloped countries (Kuzu & Santur, 2023). Hence, as maternal poisoning and fetal deaths pose critical looming risks for both the fetus and the mother (Kuzu & Santur, 2023), the ability to accurately discern abnormal fetal heart rate (FHR) patterns has significant impacts on perinatal health and neonatal mortalities (Alfirevic et al., 2017). This paper explores and evaluates the implementation of two supervised machine learning models, namely, the use of Support Vector Machines (SVM) and artificial neural networks (ANN) in application to classify fetal health conditions with cardiotocography data and detected FHR patterns.

ANNs define a broad range of neural networks that consist of an input layer, an output layer, and at least one hidden layer (Nandipati & XinYing, 2020). The use of multiple hidden layers can capture complex relationships in the data however it could become computationally expensive as complexity increases. SVM classifiers use kernel functions to construct a maximized separation hyperplane with support vectors between data classes to perform classification (Nandipati & XinYing, 2020). It is chosen in our case for advantages when handling high dimensional input data, however, extending SVMs beyond binary classification problems requires the use of one-vs-one or one-vs-rest decision functions.

Based on findings from Kuzu & Santur (2023), it can be expected for both models to achieve high performance in terms of classifying fetal health, however we expect the ANN to excel in our case due to its ability to capture complex nonlinear patterns in the data.

DATA

A cardiotocograph (CTG) records the heart rate of the fetus together with uterine contractions of the mother which can influence changes in FHR. A measured/established FHR baseline is used to monitor periods of transient increases or decreases in fetal heart rate known as acceleration or deceleration (Alfirevic et al., 2017). The Cardiotocography dataset (CTG) from the UCI Machine Learning Repository contains clinical measurements of fetal health extracted from cardiotocograms (Campos & Bernardes, 2000). The dataset has 2126 instances over 21 features with a target variable Y (fetal health) of three classes (normal: 0, suspected: 1, pathological: 2). While the dataset was well maintained with no missing values, extensive data wrangling and normalization were required. Feature re-scaling and normalization were undertaken with special considerations for the underlying medical

context of each feature and differences in measurement scales. Z-score transformation was used for normalizing specific features from extracted cardiotocogram measurements of fetal heart rate (FHR), however, a min-max scaler was applied to other features such as clinical baselines where it did not make sense for these variables to have negative values. Furthermore, percentage measures (out of 100) were also re-scaled as decimals. Finally, the target variable was encoded to 0, 1, and 2 respectively (N, S, P).

This dataset has imbalanced target classes, with the normal class (N) as the majority with approximately 77%, and minority classes susceptible (S) and pathological (P) with 13.87% and 8.28%, respectively. Imbalanced data is a common pitfall of the medical risk evaluation domain (Zieba et al., 2014) and addressing it properly in our case is vital to the integrity of our model. With this in mind, we apply stratified sampling techniques for data partitioning and incorporate adaptive synthetic sampling (AdaSyn) for modeling purposes in our methodology to remedy this issue.

Table 1. Clinical Features

Feature	Description	Feature	Description
NSP	Fetal State Class	ASTV	Abnormal Variability (% time)
LB	Baseline FHR (bpm)	MSTV	Mean Short Term Variability (% time)
AC	FHR Accelerations (/s)	ALTV	Abnormal Long Term Variability (% time)
FM	Fetal Movements (/s)	MLTV	Mean Long Term Variability (% time)
UC	Uterine Contractions (/s)	Width	Width of FHR histogram
DL	Light Decelerations (/s)	Min	Minimum of FHR histogram
DS	Severe Decelerations (/s)	Max	Maximum of FHR histogram
DP	Prolonged Decelerations (/s)	Tendency	FHR histogram Symmetry
Nmax	Number of FHR histogram peaks	Mean	Mean of FHR histogram
Nzeros	Number of FHR histogram zeros	Mode	Mode of FHR histogram
Var	Variance of FHR histogram	Median	Median of FHR histogram

METHODOLOGY

The CTG dataset was partitioned into training and test sets with an 80/20 ratio split using stratified sampling to retain the proportions of imbalanced classes within the final test set in order to preserve the underlying distribution of classes in the real world. The training set was then balanced using AdaSyn to generate synthetic samples for the two minority classes where synthetic samples are generated to match the number of instances of the majority class. Given the highly imbalanced nature of the CTG data, the AdaSyn algorithm was chosen over other popular sampling techniques like SMOTE due to the key difference that it uses a density distribution based criterion that adjusts to the underlying data distribution and can focus on difficult-to-learn examples by autonomously shifting classifier decision boundaries where there is high class overlap (Haibo He et al., 2008). The class-balanced training set was extended to contain 3949 instances under the AdaSyn algorithm which is then used for model tuning and validation.

Pre-modeling feature selection was retrospectively incorporated into the methodology to improve the intermediate modeling results. Initial data analysis revealed that some features presented multicollinearity issues. Specifically, centric measures of the cardiotocograph histogram such as mean, median, and mode were highly correlated and redundant. Nandipati and XinYing (2020) demonstrated in their research that feature selection approaches lead to improved accuracy and performance for machine learning based classifiers using the same CTG dataset. Therefore, we used recursive feature

elimination to iteratively backward eliminate features based on the Variance Inflation Factor using a threshold of 10. This reduced the number of features from the original 21 to a subset of 16 features.

K-Fold cross validation was used for hyper-parameter tuning, model training, and validation. With considerations to the number of folds (k), the size of each fold, as well as the training size of $k - 1$ folds, $K = 6$ was chosen to strike a good balance between the bias and variance trade-off with respect to the size of our synthetically balanced dataset.

Hyperparameter tuning was employed using cross-validated grid search following a two-stage procedure — architectural hyperparameter tuning and optimization hyperparameter tuning. The architectural hyperparameter tuning was first conducted on each of the models to determine the most optimal model structure (e.g., choice of decision kernels for SVM, number of hidden layers for ANN, etc.) followed by a secondary gridsearch was applied after to tune for optimization hyperparameters such as momentum, weight decay, and batch optimization. The best performing models selected from hyperparameter tuning were then re-trained on the entirety of the balanced training set to obtain final model parameters and weights before being evaluated on the test set.

The final tuned models were evaluated on multiple performance metrics. With the presence of data imbalance, the accuracy metric is not enough to assess the validity of the model (Vani, 2019). Therefore, we incorporate multiple evaluation metrics including the use of the geometric mean on top of precision, recall, specificity and F1 score. The advantage of the G-Mean metric is that it accounts for both the true positive rate with sensitivity and the true negative rate with specificity (Vani, 2019). Furthermore, we evaluate our algorithms’ performance on class-wise case, macro average, and weighted (by support) average for each metric.

MODELS

Base models were constructed and trained on the imbalanced training data to gauge the degree of impact of the imbalance issue on each model. The SVM was constructed with the default linear kernel and one-verse-rest decision function, while the ANN implemented a standard Multilayer Perceptron (MLP) structure with one hidden layer and 16 hidden neurons (to match the input dimensions) and softmax activation function with a cross entropy loss as the criterion.

Table 2. Grid Search Results

ANN			SVM		
Stage	Parameter	Values	Stage	Parameter	Values
I	hidden layers	1 , 2, 3	I	kernel	linear, radial , poly, sigmoid
I	hidden dim	4, 6, 8, 10 , 12, 14, 16	I	gamma	scale , auto, 0.1, 0.01, 0.001
II	learning rate	0.1 , 0.01, 0.001	I	degree	2, 3, 4, 5
II	dropout	0, 0.2 , 0.4, 0.5, 0.8	II	tolerance	1e-2, 1e-3 , 1e-4
II	optimizer	SGD , Adam, Rprop	II	C	0.01, 0.1, 1, 10, 100
II	weight decay	0 , 1e-2, 1e-3, 1e-4		—	—
II	momentum	0, 0.25, 0.50, 0.75, 0.85, 0.95		—	—
II	batch size	32, 64, 96, 128		—	—

ANN Model – Architectural hyper-parameter tuning iterated through different model structures by sequentially adding up to 3 hidden layers with various hidden dimensions. The optimal architecture determined via gridsearch was a single hidden layer with 10 hidden neurons. Building upon this model,

the second stage gridsearch identified an optimal learning rate of 0.01 and a dropout of 20% in the hidden layer. Stochastic gradient descent was used with momentum of 0.95, a weight decay of 0, and a batch size of 128. The final model also implemented the use of early stopping and a learning rate scheduler to adjust the learning rate when the validation loss metric was no longer improving.

SVM Model – Following the same methodology as the ANN model, the best SVM model structure was identified to have a radial basis function with the default ‘scaled’ gamma parameter. Furthermore, a regularization term of $C = 100$ and a tolerance of 0.001 was selected through the grid search. Detailed grid search parameters and results are provided in table 2.

RESULTS

Evaluation results of the base models (Appendix A4) trained on the imbalanced data revealed that the SVM was more resilient to the data imbalance, while the ANN only predicted the majority class. Intermediate modeling results (Appendix A5) showed that hyperparameter optimization after stage II significantly improved both models.

In the final models, while the SVM model had a higher overall accuracy (92%) compared to the ANN model at 88%, class-wise results reveal that both models are significantly weaker in predicting the suspected class (S) compared to the other classes. This can be attributed to the underlying characteristics of the S class — they exhibit enough variability from the normal class, yet not to the extent of the pathological class (P) making it more difficult for the model to distinguish. Furthermore, from a class-wise perspective, we see from table 3 that the susceptible class (S) is the weakest in terms of evaluation metrics for both models. This is likely due to the inherent characteristics of the susceptible class which exhibits variability from normal FHR but not fully pathological.

Table 3. Model Results

Class	ANN					SVM					Support
	Precision	Recall	Specificity	F1 Score	G Mean	Precision	Recall	Specificity	F1 Score	G Mean	
N	0.99	0.89	0.96	0.94	0.92	0.96	0.96	0.84	0.96	0.90	322
S	0.59	0.85	0.90	0.69	0.88	0.72	0.69	0.96	0.71	0.82	59
P	0.76	0.89	0.97	0.82	0.93	0.86	0.86	0.99	0.86	0.92	35
M AVG	0.78	0.87	0.82	0.82	0.91	0.84	0.84	0.93	0.84	0.88	/426
W AVG	0.91	0.88	0.95	0.89	0.92	0.91	0.92	0.87	0.91	0.89	/426

It is evident from table 3 that SVM performed better in terms of F1 scores both on a macro and weighted scale across three classes indicating a better precision and recall trade-offs, while ANN is superior in terms of geometric mean which means it is better balanced in terms of sensitivity and specificity leading to less bias towards a specific class.

From the ROC/AUC graph that displays one-vs-rest class-wise AUC for both models, we can see that both ANN and SVM excel in identifying the majority class (N) with 0.9742 and 0.9683 AUC respectively. This is also true for the P class where the same AUC (0.9782) was achieved between the two models, however, the ANN model is significantly better in capturing the S class (0.9366 vs 0.8627). Overall, the ANN model has both higher macro and micro AUC due to the SVM having low class-wise AUC for class S.

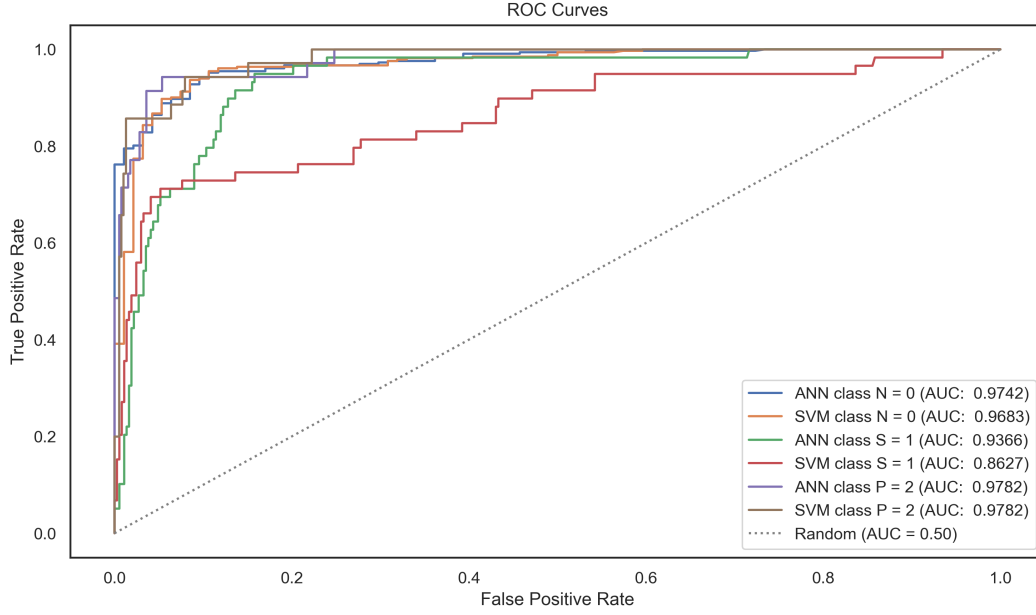


Figure 1: Class-Wise Receiver Operator Characteristics

Confusion Matrix for ANN Final Model				Confusion Matrix for SVM Final Model					
		Predicted Values					Predicted Values		
		normal	suspected	pathological			normal	suspected	pathological
Actual Values	normal	296	33	3	Actual Values	normal	319	12	1
	suspected	4	46	9		suspected	14	41	4
	pathological	2	4	29		pathological	1	4	30

Figure 2: Confusion Matrices

With the context of medical decisions in mind, false positives in classifying a healthy fetus state (N) as unhealthy (S or P) poses a much less serious medical risk for the fetus compared to misclassifying an unhealthy fetal state as healthy (false negative), which can lead to risks of serious medical complications in missing abnormal FHR patterns. Sensitivity (recall) for the unhealthy classes (S and P) should be prioritized as higher sensitivity directly indicates lower false negative rates. The ANN model excels in class-wise recall measures for both class S (0.85) and class P (0.89) compared to the SVM model (0.69 and 0.86 respectively). Secondly, the ANN model is also superior in terms of class-wise geometric mean, and is dependent on sensitivity.

CONCLUSION

Our research finds that while Support Vector Machines obtained a higher overall F1 measure compared to the artificial neural network, the ANN is preferred as it has a lower false negative rate when

classifying minority classes and a less class-wise bias by achieving a higher geometric mean.

While both models excel in predicting the majority class, the tolerance towards misclassifying the minority class remains high in the medical domain, therefore, both models should be further explored to produce more acceptable results for medical practices without overfitting.

The data imbalance issue is a limiting factor in our research as synthetic oversampling is only as good as the available data's underlying distribution. Therefore, if the distribution of minority samples within the dataset is not representative of the overall population (unhealthy fetal states), the generalization of the models to other fetal datasets can be inaccurate.

Future work includes employing the use of ensemble methods (Kuzu & Santur, 2023) for SVM classifiers and adding more extending the ANN to deep learning approaches as seen in research from Vani (2019). Furthermore, to address the data imbalance beyond the data level and apply balanced learning techniques at the algorithm level during the training process (Zieba et al., 2014) may help further improve the results of both ANN and SVM models.

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APPENDIX

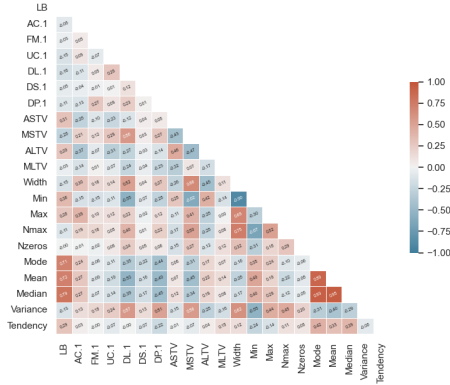


Figure A1: Pearson Correlation Matrix

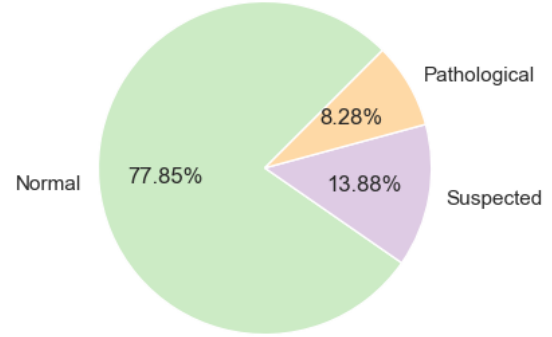


Figure A2: Class Imbalance Proportions

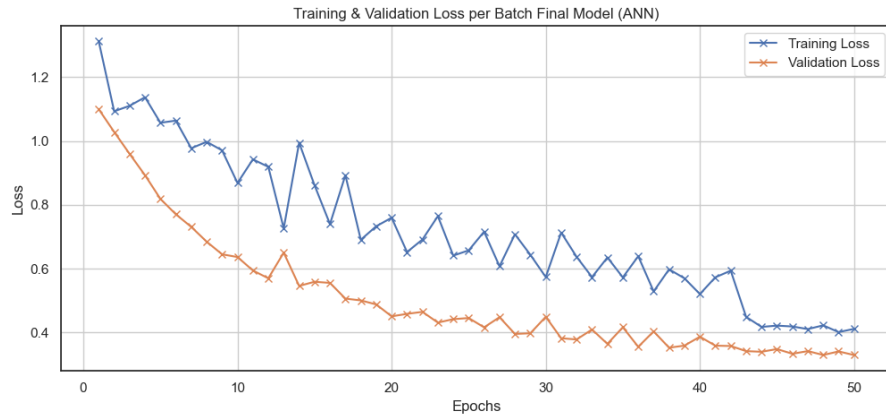


Figure A3: ANN Training and Validation Loss

Table A4. Intermediate Model Results

Class	ANN			SVM			Support
	Precision	Recall	F1 Score	Precision	Recall	F1 Score	
N	0.78	1.00	0.88	0.93	0.95	0.94	322
S	0.00	0.00	0.00	0.65	0.61	0.63	59
P	0.00	0.00	0.00	0.79	0.66	0.72	35
ACC	—	—	0.78	—	—	0.88	/426
M AVG	0.78	0.87	0.82	0.84	0.84	0.84	/426
W AVG	0.91	0.88	0.89	0.91	0.92	0.91	/426

Table A5. Intermediate ANN Results w/o Hyperparameter Optimization

Class	Precision	Recall	Specificity	F1 Score	G Mean	IBA Index	Support
N	0.91	0.33	0.88	0.49	0.54	0.28	322
S	0.00	0.00	1.00	0.00	0.00	0.00	59
P	0.11	0.97	0.31	0.20	0.55	0.32	35
W AVG	0.72	0.34	0.85	0.39	0.47	0.24	/426

Table A6. Intermediate SVM Results

Class	Precision	Recall	Specificity	F1 Score	G Mean	IBA Index	Support
N	0.96	0.91	0.88	0.94	0.90	0.81	322
S	0.59	0.75	0.92	0.66	0.83	0.67	59
P	0.81	0.86	0.98	0.83	0.92	0.83	35
W AVG	0.90	0.88	0.90	0.89	0.89	0.79	/426

Tbl. A4 presents the baseline results for both the ANN and SVM models. We see that the SVM initially outperforms the ANN as it is more resilient to class imbalance. The ANN model, in contrast, failed to classify minority classes and favoured heavily in predicting the majority class. As seen in **tbl.** the ANN model shows low performance prior to optimization hypertuning as it did not classify the S class. Intermediate results of the SVM model showed significant improvement in class P performance compared to the base model (**tbl.** A6).